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|---|-----|-------|---|----------|-----|
| 7 | (a) | (i) | <i>either</i> helium nucleus
or particle containing two protons and two neutrons | B1 | [1] |
| | | (ii) | allow any value between 1 cm and 10 cm | B1 | [1] |
| | (b) | (i) | energy = $(8.5 \times 10^{-13}) / (1.6 \times 10^{-13})$
= 5.3 MeV | M1
A0 | [1] |
| | | (ii) | number = $(5.3 \times 10^6) / 31$
= 1.7×10^5 (<i>allow 2 s.f. only</i>) | C1
A1 | [2] |
| | | (iii) | number per unit length = $(1.7 \times 10^5) / (a)(ii)$
correct numerical value
correct unit | A1
B1 | [2] |